



Summary

Reference: RA127799/1

Sign-off Status: Authorised

Date Created:	18/03/2026	Confidential?	No
Assessment Title:	Manufacturing Extractor Prototype for MECH0073		
Assessment Outline:	Manufacturing for MECH0073 prototype is expected to be done at our master site MechSpace and at our industry partner's site at Agfa Cambridge. This risk assessment is primarily for manufacturing work done at Agfa Cambridge, as there is already a risk assessment for the project at MechSpace		
Area Responsible (for management of risks)	Location of Risks		
Division, School, Faculty, Institute:	Faculty of Engineering Sciences	Building:	Off-Site
Department:	Dept of Mechanical Engineering	Area:	
Group/Unit:	All Groups/Units	Sub Area:	
Further Location Information:	515 Coldhams Lane, Cambridge, CB1 3JS Agfa 44 Wicklow St, London WC1X 9HL MechSpace		
Is additional GM or HG approval required? Only relevant to specialist biological risk assessments (GMM2, GMM3, HG2, HG3, GM animals, GM plants) except GMM class 1.:	Click SELECT to change <u>ONLY</u> if this is a GMM Class 2, GMM Class 3, HG2, HG3, GM animals or GM plants risk assessment		
Assessment Start Date:	19/03/2026	Review or End Date:	26/11/2026
Country Location:	GMM Class 2, GMM Class 3 or Hazard Group 3 Risk Assessment :		
Relevant Attachments:	Description of attachments: Location of non-electronic documents: MechSpace risk assessment is attached to our project boxes labelled B07-08		
Assessor(s):	Goh, Teik Beng		
Approver(s):	Jose Castrejon Pita		
Signed Off:	Jose Castrejon Pita (19/03/2026 11:44)		

PEOPLE AT RISK (from the Activities covered by this Risk Assessment) *

CATEGORY
Employees
Undergraduates
Other

Who Made By	Date/Time Received	Comments
Castrejon Pita, Jose	19/03/2026	After assessment, the rig will remain in UCL East, the Marshgate building. It will eventually move to Agfa.

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1. General Site Hazards at Agfa Cambridge

Description of Activity:	Site hazards at industry partner's site
List those managing this Activity and their competence:	

Hazard 1. Injury at site

Certain parts that we designed are expected to be machined with the assistance of our industry partner at their site	Existing Control Measures Instructions and safety briefs to be followed. Team to remain vigilant and aware of the environment during the visit. Suitable safety equipment and wear to be used during the visit, e.g. safety glasses, if required by the industry partner. Students and staff to follow any safety operating procedures. Students have also been to the site multiple times prior for technical visits and meetings. Staff to brief the students about expected standards and code of behaviour. Identify safety requirements at time of arrival. Identify fire exits.	Further Control Measures
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Hazard 2. Slips, Trip and Falls

Trips over objects around partner workshop.

Existing Control Measures

Partner workshop area is generally kept organised and tidy, and well organised between pathways and working areas.

Area is well lit.

Students and staff are to remain spatially aware of their surroundings at all times.

Further Control Measures

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Hazard 3. Manual handling of prototype components

Strain from lifting prototype components

Existing Control Measures

Components are mostly below 5kg.

Heavier machined parts are done at the site of industry partner.

Transport of components and equipment are shared equally between students.

Lifting aids used where available.

Moving heavier objects at site assisted by industry partner when necessary.

Brief correct lifting technique; avoid twisting while carrying; plan rest points if moving long distances

Further Control Measures

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Risk Level

With Existing Controls:

Risk Level **A - Very Low / Trivial** Likelihood Severity

2. Manufacturing of Prototype

Description of Activity: Manufacturing of prototype involves 3D-printing, machining, and some electrical work for connecting sensors and display to the prototype.

List those managing this Activity and their competence:

Hazard 1. 3D Printing Emissions

Emission of ultrafine particles and volatile compounds from heated filament/resin during printing, potentially causing respiratory or eye/airway irritation, especially in poorly ventilated rooms.

Existing Control Measures

Most 3D-printing is done at MechSpace, which has a dedicated print room.

There is a possibility that some jobs may happen at the partner site, which is safe from emissions as the manufacturing/workshop area is large and well-ventilated

Further Control Measures

Hazard 2. Soldering

Contact burns from heated solder iron.

Soldering is expected to be used when assembling the sensors to their respective breadbox/Arduino circuitry to the extractor.

Existing Control Measures

Soldering work is done at dedicated soldering tables either at MechSpace or at the partner site.

Appropriate PPE worn when soldering.

At either location, guidance will be sought if there is any doubt in the process.

Further Control Measures

Hazard 3. Hot surfaces and burns

3D print build plates, freshly machined parts and solder irons are hot and may cause burns upon touch.

Existing Control Measures

Allow for parts to cool down before handling.

Wear gloves when soldering.

Further Control Measures

Hazard 4. Use of resins and adhesives

Some components are large and are unable to be printed as a single body, hence we plan to use epoxy resin or possibly a PLA glue to join these parts together.

These adhesives can cause irritability and also bind skin together upon contact.

Existing Control Measures

Wear gloves while using appropriate tools when applying adhesive and while handling for curing process.

Further Control Measures

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Hazard 5. Cuts and scrapes

Possibility of getting minor cuts and scrapes while handling parts, such as from removing supports/rafts from 3D prints, or while assembling/disassembling prototype.

Existing Control Measures

Gloves used if necessary.

First-aid kits available both at MechSpace and at partner site.

Further Control Measures

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Hazard 6. Machining

The main housing frame of our extractor prototype is to be machined.

Existing Control Measures

Machining is done at our partner site, and done by their technicians, who are accustomed to machining parts in their own work.

Further Control Measures

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Risk Level

With Existing Controls:

Risk Level **B - Low / Tolerable** Likelihood Severity

Actions

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Actions associated with this Risk Assessment	
*** No Actions have been recorded***	